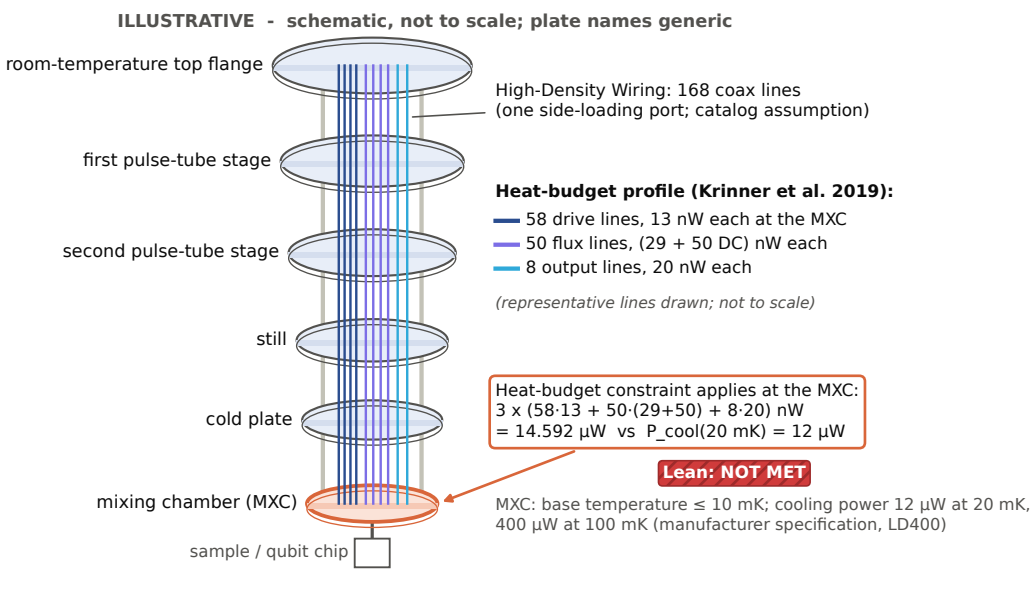
ILLUSTRATIVE - schematic, not to scale; plate names generic
room-temperature top flange
first pulse-tube stage
second pulse-tube stage
still
cold plate
mixing chamber (MXC)
sample / qubit chip
High-Density Wiring: 168 coax lines
(one side-loading port; catalog assumption)
Heat-budget profile (Krinner et al. 2019):
58 drive lines, 13 nW each at the MXC
50 flux lines, (29 + 50 DC) nW each
8 output lines, 20 nW each
(representative lines drawn; not to scale)
Heat-budget constraint applies at the MXC:
3 x (58·13 + 50·(29+50) + 8·20) nW
= 14.592 µW vs P_cool(20 mK) = 12 µW
Lean: NOT MET
MXC: base temperature ≤ 10 mK; cooling power 12 µW at 20 mK,
400 µW at 100 mK (manufacturer specification, LD400)